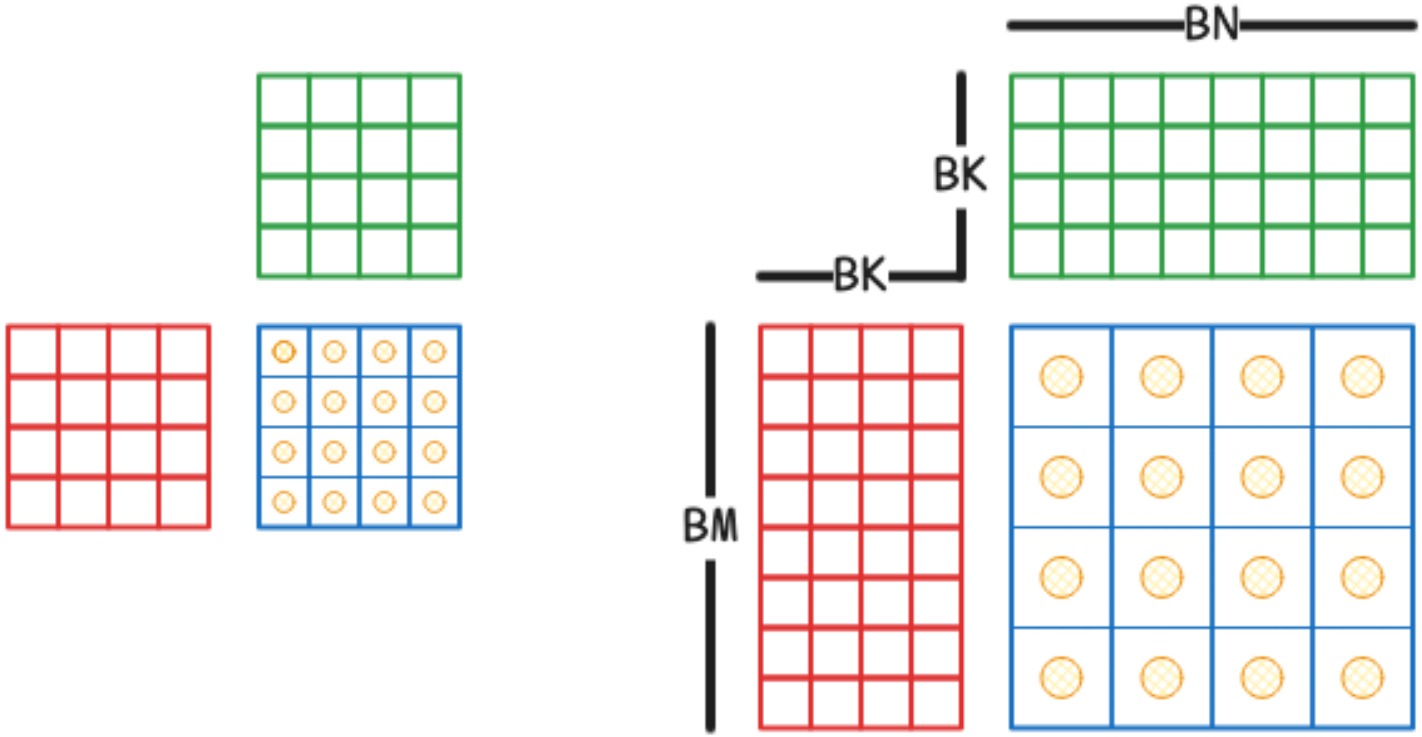
Section Overview

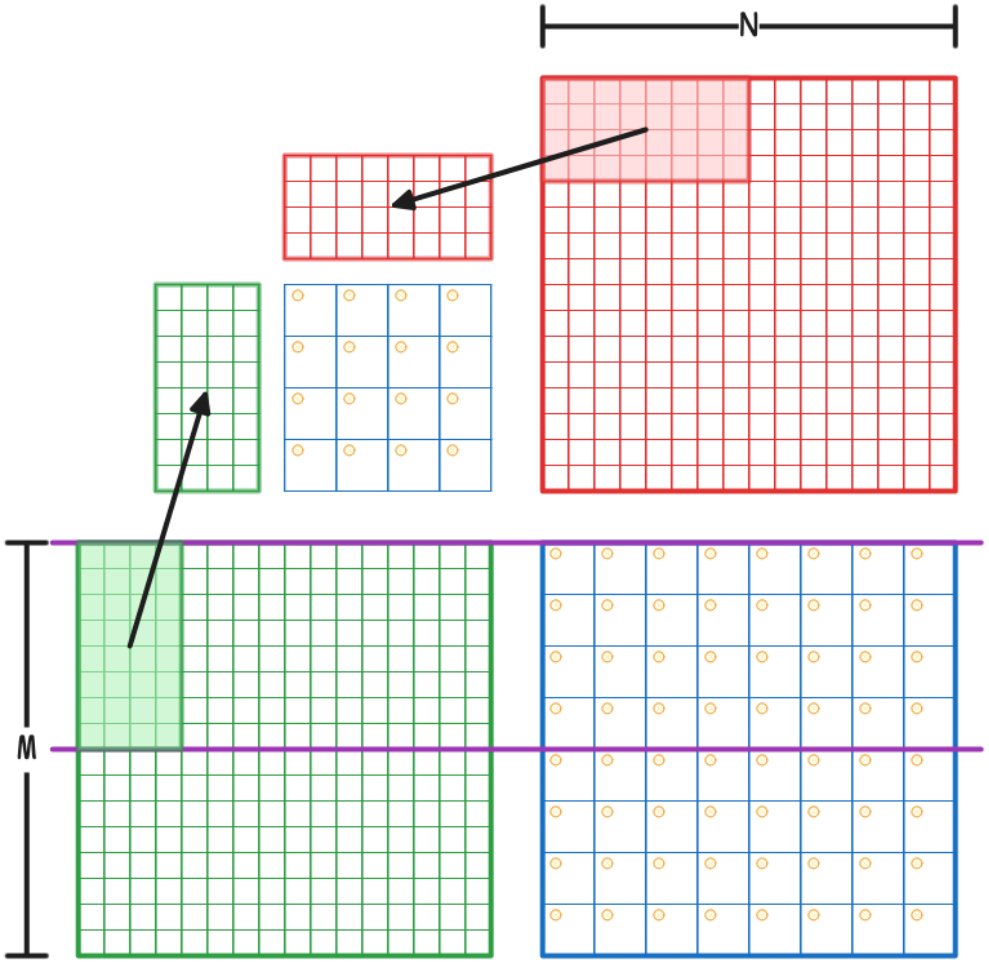
- Applying the Roofline Model: Block Tiling and Thread Coarsening

Thread coarsening: Increasing AI

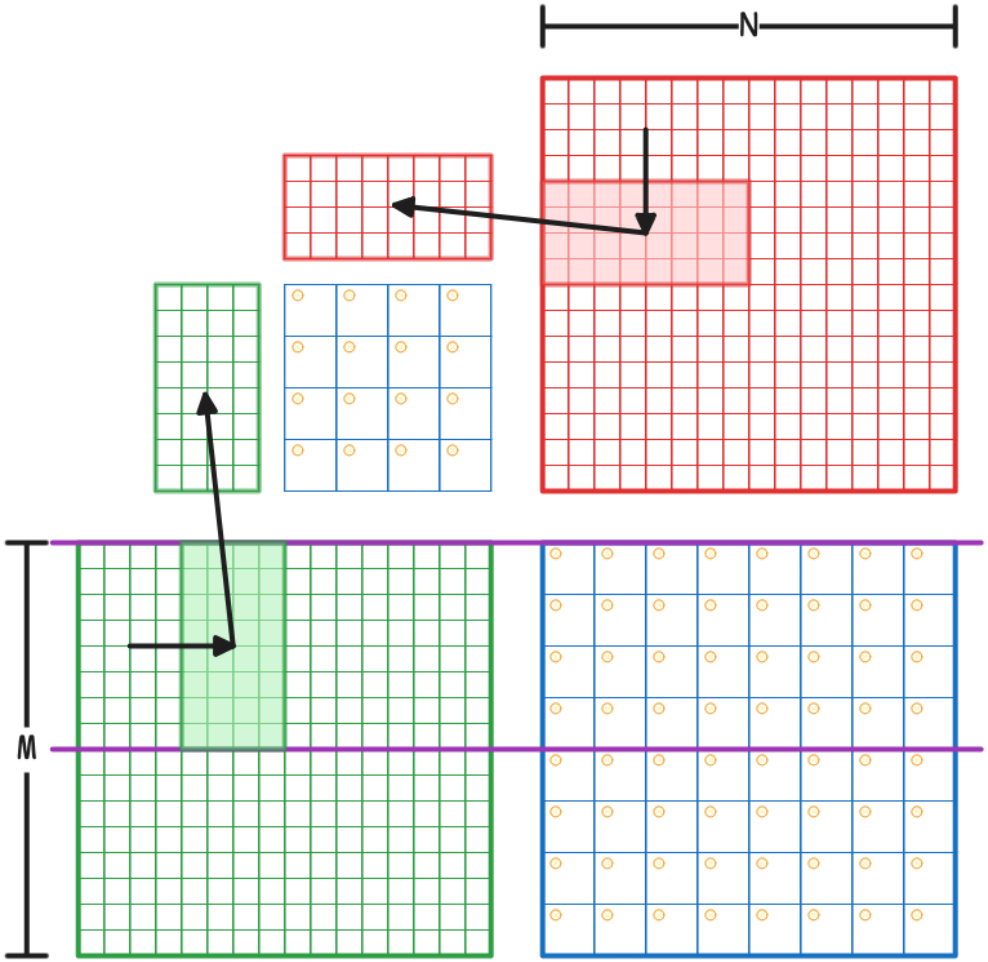
- Basic tiling assigns 1 element of the output to each thread.
- We can *coarsen* each thread, and assign a $TM \times TN$ portion of the output to each thread.
- We load larger chunks into SMEM, increasing data reuse and AI.
- One loaded A value can update several columns; one loaded B value can update several rows.



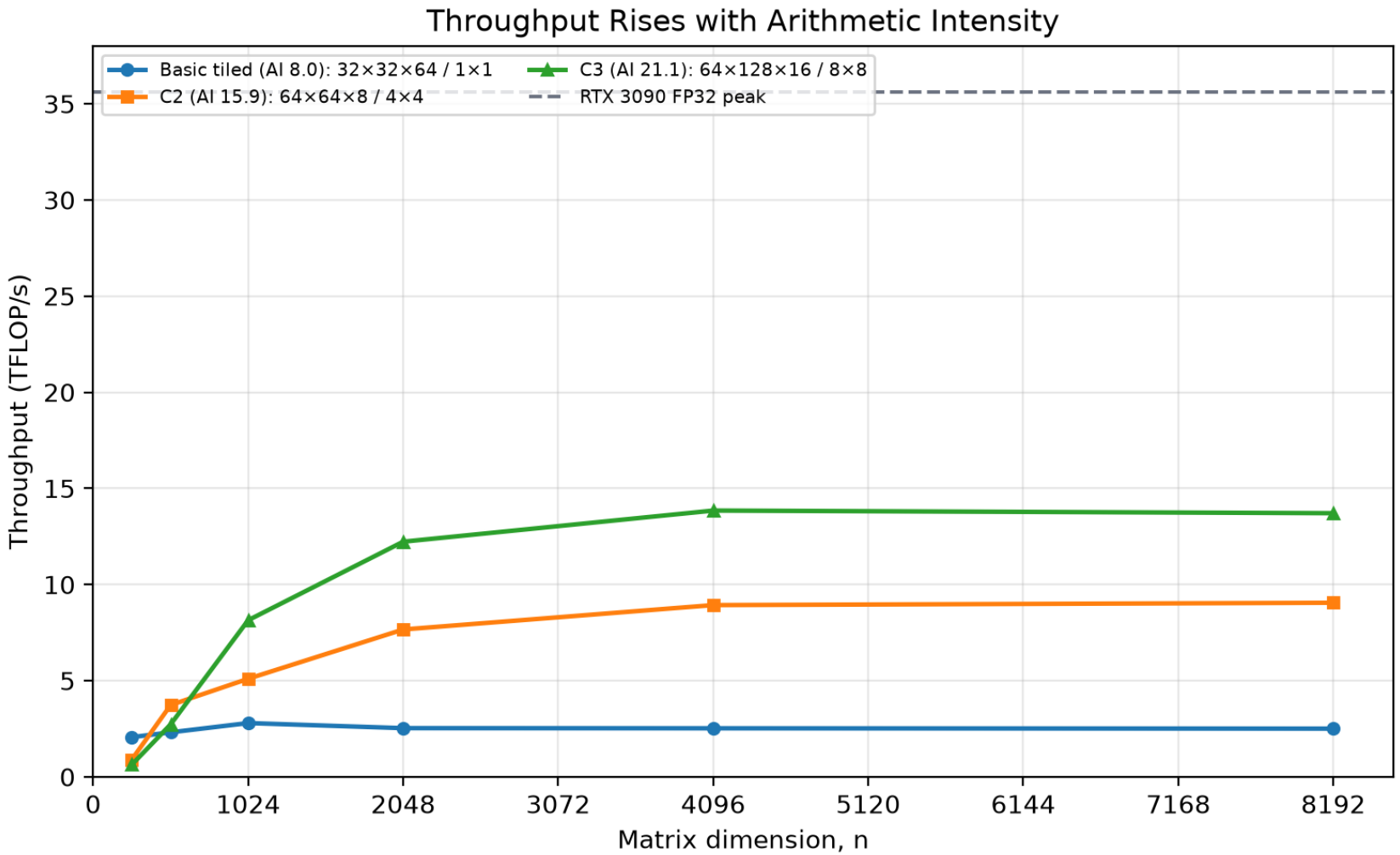
Coarsened SGEMM visualization



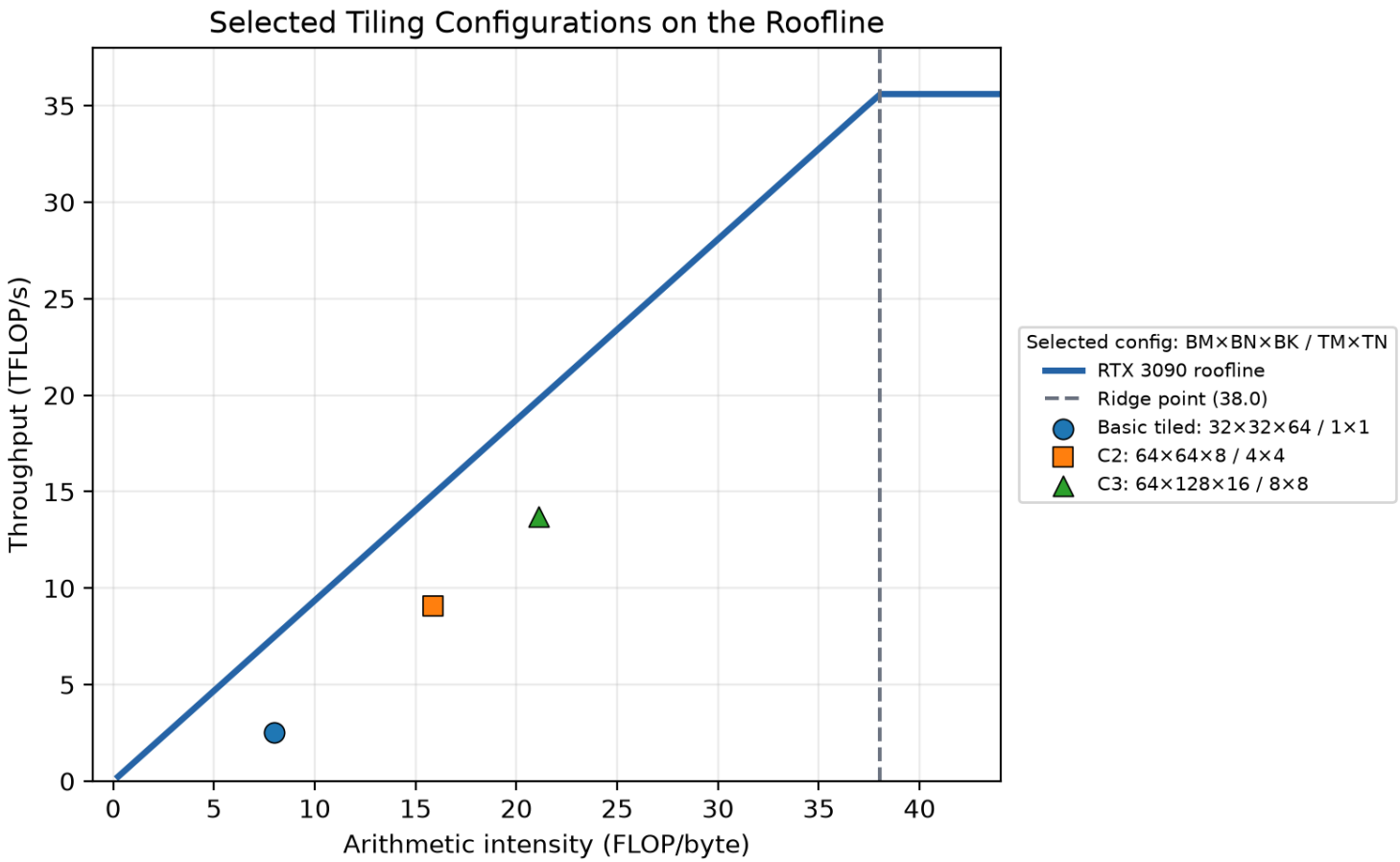
Coarsened SGEMM visualization



Higher-AI tiles tend to raise throughput

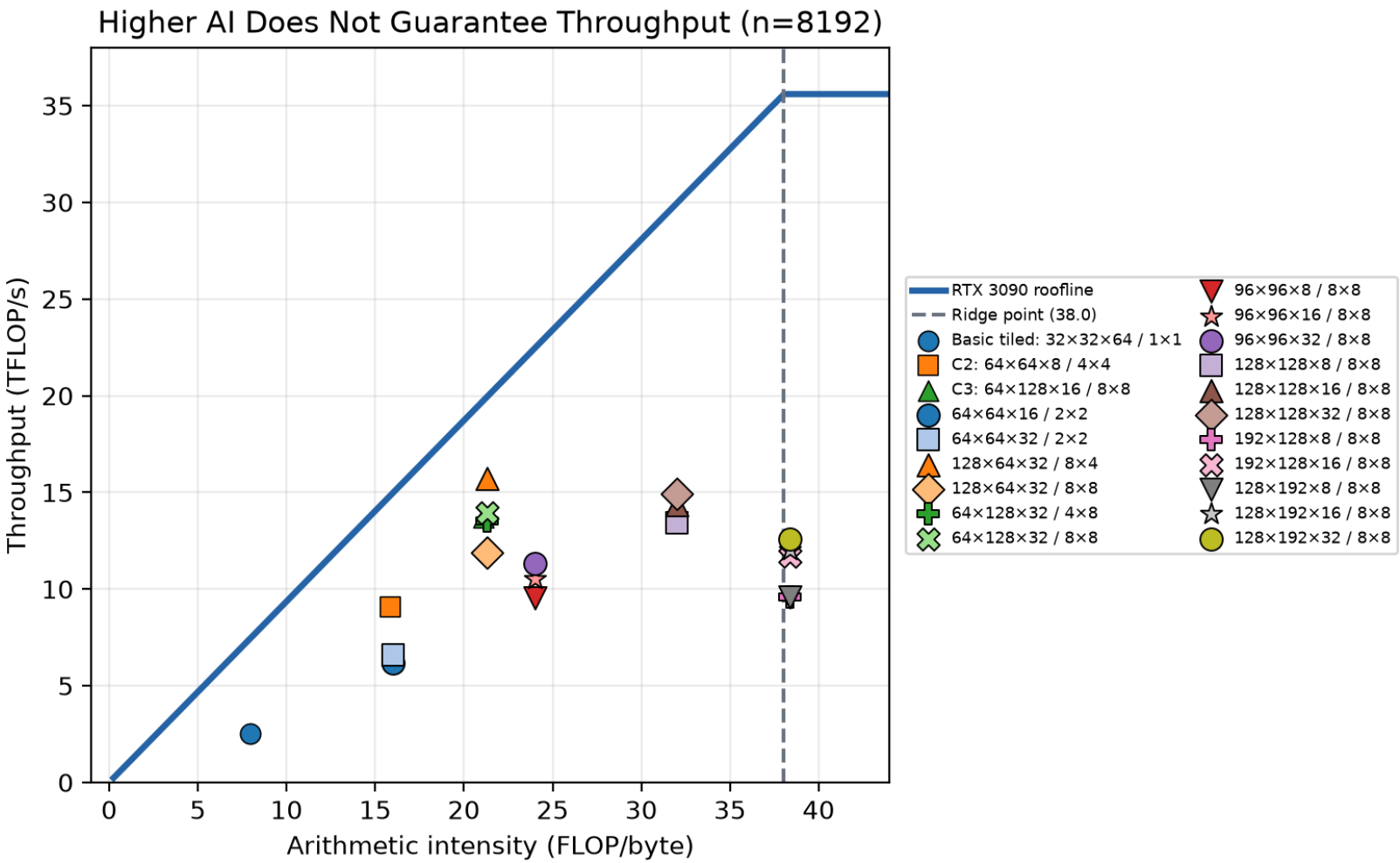


Tiling configurations on the roofline



Reminder that the Roofline Model only tells us the upper bound!

Higher AI does not guarantee higher throughput



The roofline is an upper bound, not a performance prediction.

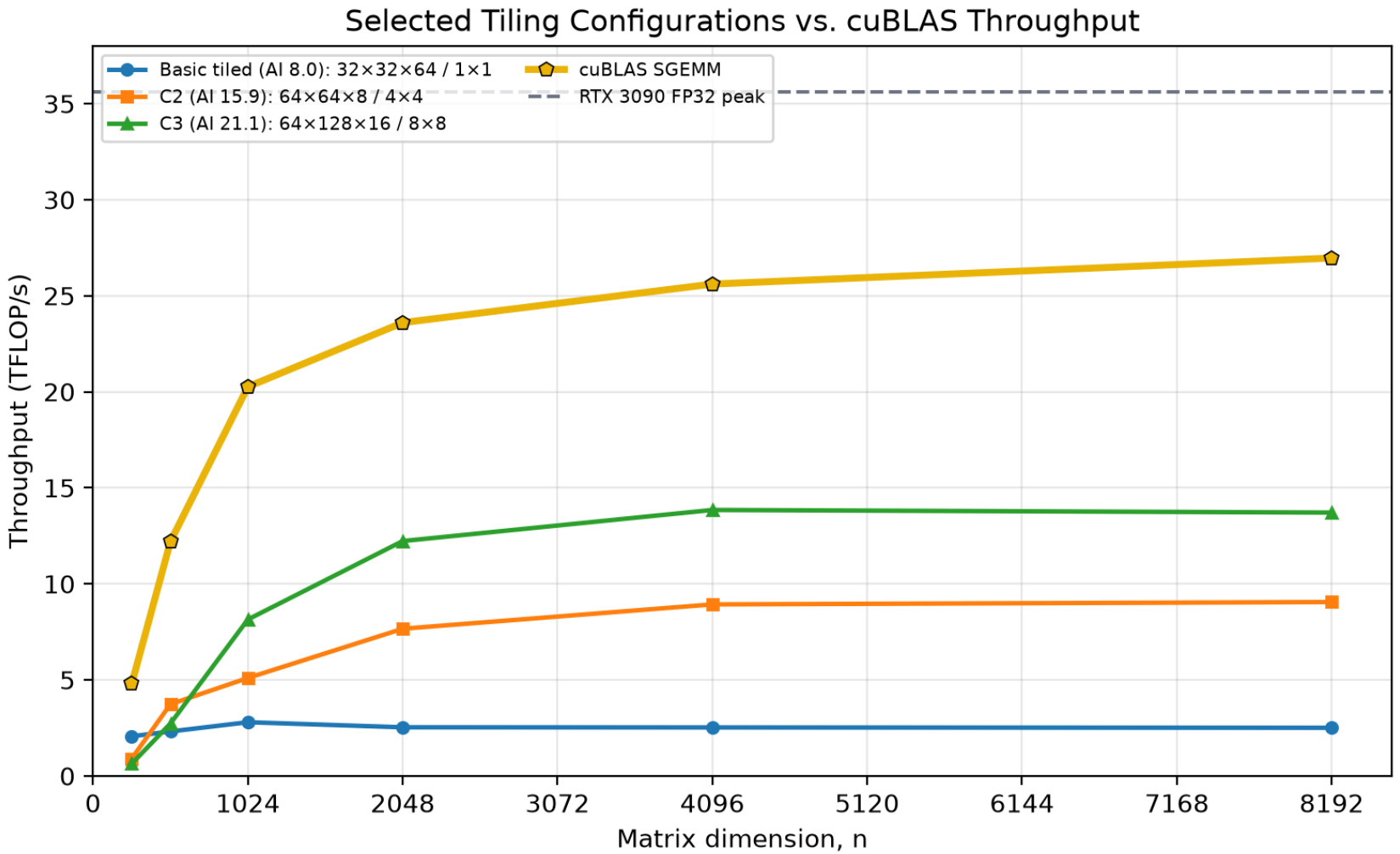
We have only scratched the surface: cuBLAS

- cuBLAS is NVIDIA's architecture-tuned linear algebra library and the practical performance target for our SGEMM kernels.
- At $n = 8192$, cuBLAS reaches 26.96 TFLOP/s, while the highest-AI selected configuration (C3) reaches 13.71 TFLOP/s, or 50.8% of cuBLAS's throughput.
- cuBLAS completes the multiplication in 40.8 ms versus 80.2 ms for C3, about 1.97× faster.

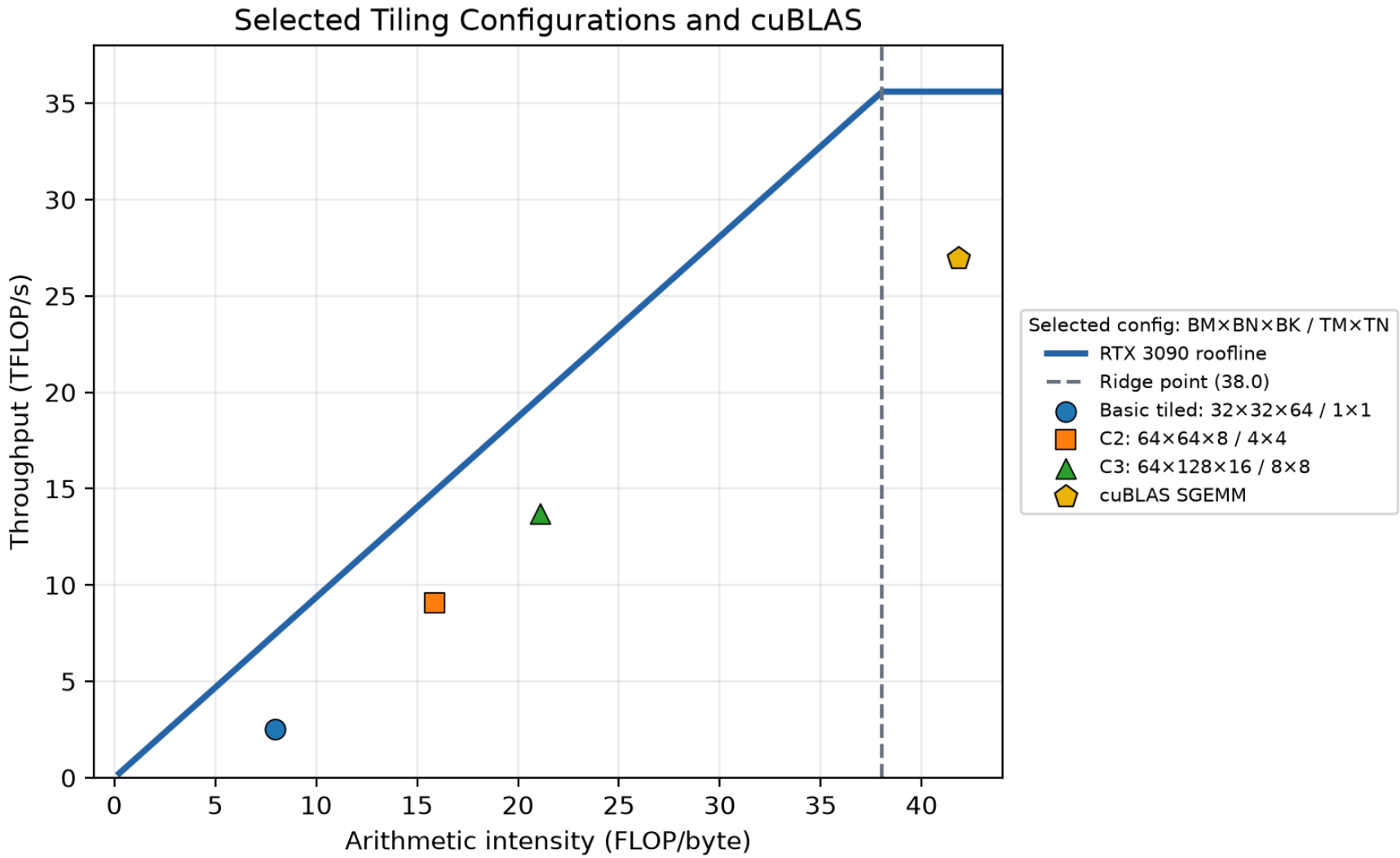
The remaining gap

Production kernels combine deeper register and warp tiling, vectorized memory access, pipelined loads, and architecture-specific instruction scheduling.

cuBLAS remains the performance target



cuBLAS on the roofline



cuBLAS arithmetic intensity is estimated from the profiled 256×128 output-tile shape.

We've come a long way

